

SONiX 32-Bit Cortex-M0 Micro-Controller

PMU Example Code

SONiX reserves the right to make change without further notice to any products herein to improve reliability, function or design. SONiX does not assume any liability arising out of the application or use of any product or circuit described herein; neither does it convey any license under its patent rights nor the rights of others. SONiX products are not designed, intended, or authorized for use as components in systems intended, for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the SONiX product could create a situation where personal injury or death may occur. Should Buyer purchase or use SONiX products for any such unintended or unauthorized application. Buyer shall indemnify and hold SONiX and its officers, employees, subsidiaries, affiliates and distributors harmless against all claims, cost, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use even if such claim alleges that SONiX was negligent regarding the design or manufacture of the part.



AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

Table of Content

AMENDMENT HISTORY	2
1 OVERVIEW.....	4
2 FEATURE	4
3 HW	4
4 EXAMPLE CODE.....	5
4.1 SLEEP MODE.....	5
4.2 DEEP SLEEP MODE	6
4.3 DEEP POWER DOWN MODE	7
5 RESULT	9
6 APPENDIX.....	9

1 OVERVIEW

This document provides our customers the C program examples of PMU.

2 FEATURE

- Sleep mode
- Deep sleep mode
- Deep power down mode
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Sleep Mode

1. Setup wakeup source in main function.
2. Call PMU_Setting(PMU_SLEEP) in main function, and then MCU enters Sleep mode.

```
//=====
//set P2.0~P2.5 as Wakeup pins (input pull-down, Rising edge trigger interrupt)
for (i=0; i<6; i++)
{
    SN_GPIO2->MODE &= ~(1<<i);          //Set as Input mode
    SN_GPIO2->CFG &= ~(0x3<<(2*i));
    SN_GPIO2->CFG |= (1<<(2*i));        //Enable Pull-down resistor
    SN_GPIO2->IEV &= ~(1<<i);          //Set as Rising edge interrupt
}
SN_GPIO2->IC = 0x3F;
SN_GPIO2->IE |= 0x3F;
NVIC_ClearPendingIRQ(P2_IRQn);
NVIC_EnableIRQ(P2_IRQn);
//=====

PMU_Setting(PMU_SLEEP);

while(1){
    SN_GPIO1->BSET = 0x1;              //Set P1.0
    UT_DelayNms(200);
    SN_GPIO1->BCLR = 0x1;              //Clear P1.0
    UT_DelayNms(200);
}
```

```
int main (void)
{
    //initWakeup source
    .
    .
    User settings
    .
    .
    PMU_Setting(PMU_SLEEP);

    .
    .
    User code
    .
    .
}
```

4.2 Deep Sleep Mode

1. Setup wakeup source in main function.
2. Call PMU_Setting(PMU_DEEP_SLEEP) in main function, and then MCU enters Deep Sleep mode.

```

61 //=====
62 //set P2.0~P2.5 as Wakeup pins (input pull-down, Rising edge trigger interrupt)
63 for (i=0; i<6; i++)
64 {
65     SN_GPIO2->MODE &= ~(1<<i);           //Set as Input mode
66     SN_GPIO2->CFG &= ~(0x3<<(2*i));
67     SN_GPIO2->CFG |= (1<<(2*i));         //Enable Pull-down resistor
68     SN_GPIO2->IEV &= ~(1<<i);           //Set as Rising edge interrupt
69 }
70 SN_GPIO2->IC = 0x3F;
71 SN_GPIO2->IE |= 0x3F;
72 NVIC_ClearPendingIRQ(P2_IRQn);
73 NVIC_EnableIRQ(P2_IRQn);
74 //=====
75
76 PMU_Setting(PMU_DEEP_SLEEP);
77
78 while(1){
79     SN_GPIO1->BSET = 0x1;           //Set P1.0
80     //P1.0 = 1;

```

```

int main (void)
{
    //initWakeup source
    .
    .
    User settings
    .
    .
    PMU_Setting(PMU_DEEP_SLEEP);

    .
    .
    User code
    .
    .
}

```

4.3 Deep Power Down mode

1. Finish the following procedures in PMU.c according to the MCU types

MCU	Procedures before entering DPD mode
Only SN32F700/SN32F100 support	1. Pull the DPDWAKEUP pin externally HIGH (Strongly recommended to set output high first, and then set as input pull-up to reduce pull-up time). 2. (Optional) Save data to be retained during Deep power-down to the DATA bits in Backup registers.

2. Call PMU_Setting(PMU_DEEP_PWR_DOWN)main function, and then MCU enters Deep power down mode.

```

73     SN_GPIO1->BCLR_b.BCLR15 = 1;           //P1.15=L
74     UT_DelayNms(200);                       //delay 200ms@IHRC=12MHz
75 }
76 //=====
77 }
78
79
80 PMU_Setting(PMU_DEEP_PWR_DOWN);
81
82 init32(0);
83 }
84

```

3. Check whether MCU are waken up from DPD mode or not in main function, and process on demand.

```

int main (void)
{
    if (SN_PMU->CTRL == PMU_DEEP_PWR_DOWN) //Wakeup from DPD mode
    {
        .
        .
        User code for DPD Wakeup
        .
        .
    }

    //initWakeup source
    .
    .
    User settings
    .
}

```

```
.  
    PMU_Setting(PMU_DEEP_PWR_DOWN);  
}
```


5 RESULT

The power consumption of MCU can meet the spec, and the MCU can be waken up.

6 APPENDIX

The debug mode is not supported in Deep-sleep and Deep Power-down mode, so users will not be able to update firmware by ICE (SN-LINK, J-LINK, and etc.) when MCU is in Deep-sleep mode or Deep power-down mode. In order to solve this issue, users can

- 1) Power off MCU.
- 2) Short BOOT pin to GND.
- 3) Power on MCU and MCU will enter Boot loader.

SONiX provides Boot loader to check the status of the BOOT pin during boot procedure. If the BOOT pin is Low during Boot procedure, MCU will execute code in Boot loader with normal mode instead of User code, so user can update firmware by ICE normally. Users can then remove the jumper which shorts BOOT pin to GND, reset MCU, and execute code after updating.

SONIX reserves the right to make change without further notice to any products herein to improve reliability, function or design. SONIX does not assume any liability arising out of the application or use of any product or circuit described herein; neither does it convey any license under its patent rights nor the rights of others. SONIX products are not designed, intended, or authorized for use as components in systems intended, for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the SONIX product could create a situation where personal injury or death may occur. Should Buyer purchase or use SONIX products for any such unintended or unauthorized application. Buyer shall indemnify and hold SONIX and its officers, employees, subsidiaries, affiliates and distributors harmless against all claims, cost, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use even if such claim alleges that SONIX was negligent regarding the design or manufacture of the part.

Main Office:

Address: 10F-1, NO. 36, Taiyuan Street, Chupei City, Hsinchu, Taiwan R.O.C.

Tel: 886-3-5600 888

Fax: 886-3-5600 889

Taipei Office:

Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.

Tel: 886-2-2759 1980

Fax: 886-2-2759 8180

Hong Kong Office:

Unit No.705, Level 7 Tower 1, Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.

Tel: 852-2723-8086

Fax: 852-2723-9179

Technical Support by Email:

Sn8fae@sonix.com.tw